

Safflower Oil Supplementation and Cardiometabolic Lipid Profiles: A Randomized, Double-Blind Investigation in Adults with Metabolic Syndrome

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Abstract

Metabolic syndrome elevates the risk of cardiovascular disease and type 2 diabetes, and dietary lipid interventions remain a low-cost, scalable strategy for managing dyslipidemia. This dissertation reports a 16-week randomized, double-blind, placebo-controlled trial evaluating high-linoleic safflower oil supplementation (8 g/day) on fasting lipid profiles in 142 adults with metabolic syndrome. Relative to placebo, the safflower group showed a statistically significant reduction in LDL cholesterol and an increase in HDL cholesterol, with no significant change in fasting triglycerides. Effects were modified by baseline insulin resistance. Findings support a modest, targeted role for safflower oil in lipid management while underscoring the limits of single-nutrient interventions.

Chapter 1: Introduction

Cardiometabolic disease is the leading contributor to global mortality, and metabolic syndrome, a clustering of central adiposity, dyslipidemia, hypertension, and impaired glucose regulation, sits upstream of much of that burden. Pharmacologic management is effective but carries cost, adherence, and side-effect challenges, which has sustained interest in dietary interventions that act on the same lipid pathways.

High-linoleic safflower oil has been proposed as a favorable dietary fat because of its polyunsaturated fatty acid content. Prior evidence is mixed, drawn largely from small or short trials with heterogeneous populations. This dissertation addresses that gap with an adequately powered, double-blind trial in a well-characterized metabolic syndrome population.

The central research question is whether 16 weeks of safflower oil supplementation produces clinically meaningful improvements in fasting lipid profiles relative to placebo, and whether baseline insulin resistance modifies the response.

Chapter 2: Literature Review

Dietary fatty acid composition influences hepatic lipoprotein metabolism, with linoleic acid associated in observational cohorts with lower LDL cholesterol. However, observational designs cannot isolate the effect of a single oil from the broader dietary pattern, and randomized evidence has been inconsistent.

Across the trials reviewed, three weaknesses recur: short duration that may miss steady-state lipid changes, inadequate blinding of an oil-based intervention, and failure to account for insulin resistance as an effect modifier. The present study was designed specifically to address these three limitations.

Chapter 3: Methods

We enrolled 142 adults (ages 30 to 65) meeting NCEP ATP III criteria for metabolic syndrome and randomized them 1:1 to 8 g/day high-linoleic safflower oil or an isocaloric high-oleic placebo oil, matched for appearance and packaging to preserve blinding. The trial ran 16 weeks with assessments at baseline, week 8, and week 16.

The primary outcome was change in fasting LDL cholesterol; secondary outcomes were HDL cholesterol, triglycerides, and HOMA-IR. Analysis followed intention-to-treat using linear mixed models

with a prespecified interaction term for baseline HOMA-IR. The study was powered at 0.85 to detect a 10 mg/dL between-group LDL difference.

Chapter 4: Results

Retention was 91 percent and groups were balanced at baseline. The safflower group showed a mean LDL reduction of 11.4 mg/dL versus 2.1 mg/dL in placebo (between-group difference significant at $p < 0.01$), and an HDL increase of 3.2 mg/dL ($p = 0.03$). Fasting triglycerides did not differ significantly between groups.

A significant interaction with baseline insulin resistance was observed: participants in the highest HOMA-IR tertile showed attenuated LDL response, suggesting insulin resistance blunts the lipid benefit.

Chapter 5: Discussion

The results support a modest but statistically significant LDL-lowering effect of safflower oil over 16 weeks, consistent with a linoleic-acid mechanism, while the null triglyceride finding cautions against overgeneralizing benefits across all lipid fractions.

The insulin-resistance interaction is the most clinically interesting finding and aligns with the hypothesis that metabolic context governs nutrient response. Limitations include a single-site sample, reliance on self-reported background diet, and a duration that, while longer than most prior work, cannot speak to long-term cardiovascular endpoints.

Chapter 6: Conclusion

Safflower oil supplementation produced a modest, targeted improvement in LDL and HDL cholesterol in adults with metabolic syndrome, modified by baseline insulin resistance. The work argues for stratifying dietary lipid interventions by metabolic phenotype rather than treating them as uniformly effective. Future research should test longer durations, multi-site populations, and hard cardiovascular outcomes.

References

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